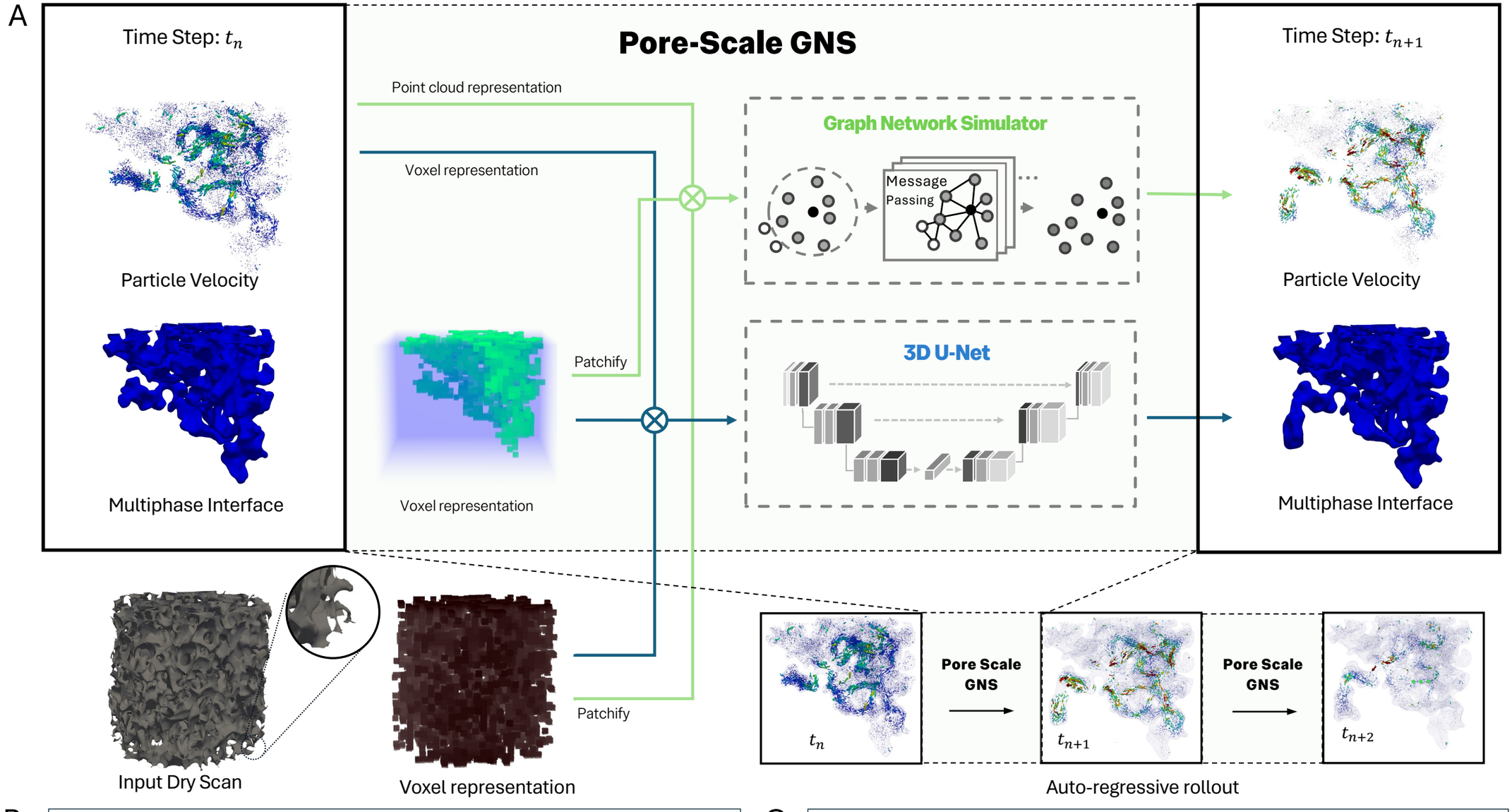
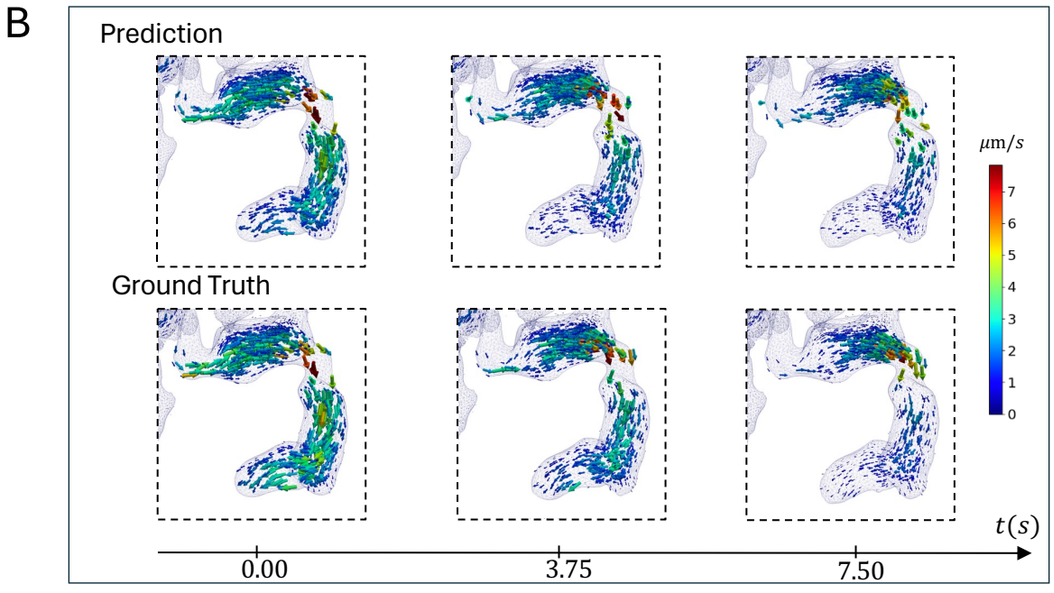
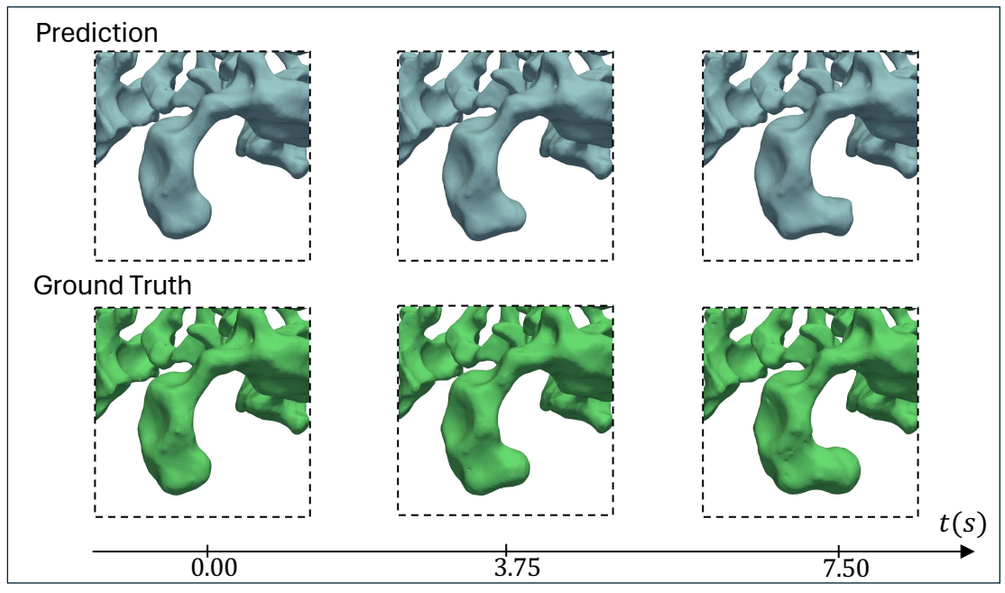

A
Time Step: t_n
Particle Velocity
Multiphase Interface
Pore-Scale GNS
Point cloud representation
Voxel representation
Graph Network Simulator
Message Passing
Patchify
3D U-Net
Voxel representation
Time Step: t_{n+1}
Particle Velocity
Multiphase Interface
Input Dry Scan
Voxel representation
Patchify
t_n
Pore Scale GNS
t_{n+1}
Pore Scale GNS
t_{n+2}
Auto-regressive rollout


B
Prediction
Ground Truth
$\mu m/s$
0
1
2
3
4
5
6
7
$t(s)$
0.00
3.75
7.50


C
Prediction
Ground Truth
$t(s)$
0.00
3.75
7.50